TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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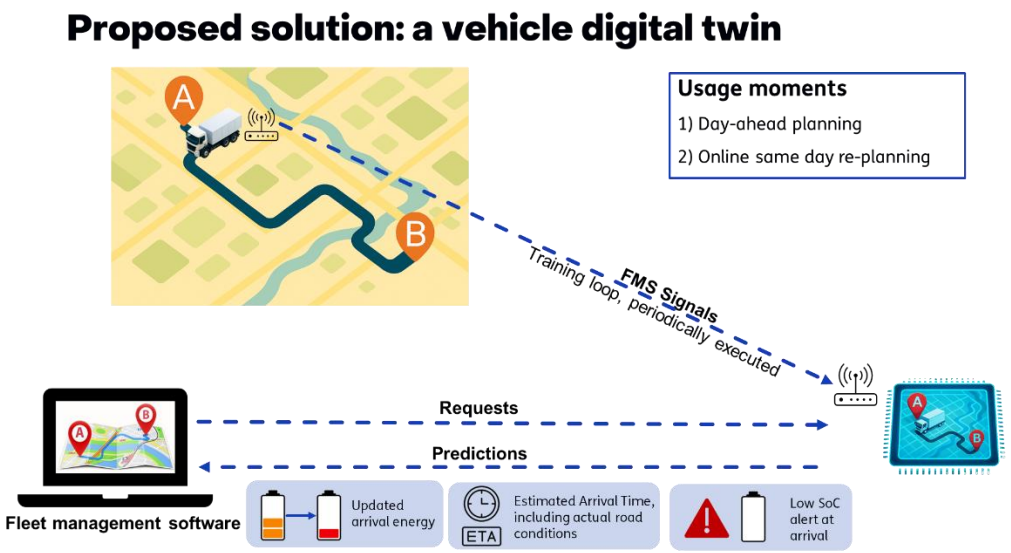
2. Technology Overview

Title of Proposed Technology	Integrated charge planning and battery digital twin platform for electric heavy-duty fleets, combining grid-aware smart charging, battery health-aware scheduling, and vehicle energy digital twins validated under megawatt-charging constraints (MCS).			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility (√), Next-generation Energy (√), Bio-health ( ), Aerospace ( ), Quantum ( ), Advanced Robotics & Manufacturing ( ), AI (√), Cybersecurity (√), Hydrogen ( ), Semiconductor materials ( )			
Industrial Technology Classification [1]	Technology Field 1 (Large Category)		Technology Field 2 (Large Category)	
	Information & Communications		Electrical & Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Software	Internet SW	Batteries	Secondary Batteries
Concept of Proposed Technology	**This proposal targets an integrated software toolchain that couples:** 1) Vehicle charge planning and grid-aware optimization, i.e. alignment with site-level grid constraints and tariffs. 2) Vehicle and battery digital twins, including charging-process prediction and battery health and degradation models.			

Core concept

- A fleet planning layer converts logistics schedules into energy demand constraints per vehicle (arrival time windows, required state of charge, route energy demand).
- A digital twin layer provides: (i) vehicle energy consumption prediction on planned routes; and (ii) battery digital twin predictions of charging power profile, charge time, and health-related constraints. Battery digital twins refer to operationally inferable charging behavior and health-aware constraints, not OEM-internal BMS models or warranty-grade battery analytics.
- An optimization layer computes a robust charging schedule under charger availability, charger-vehicle compatibility, site grid limits, and electricity price signals, while explicitly accounting for battery degradation cost terms. The fleet charging planning is a lightweight, energy-focused optimization layer developed within the project, not a prerequisite logistics system.

Heavy-duty fleets and megawatt-charging-level operating conditions are used as validation domains to ensure robustness under high power demand, tight logistics, and strict grid constraints. The resulting platform remains applicable to a broader range of grid-constrained commercial electric fleets.



TNO prior work relevant to this concept includes:

- A Charge Planning Tool (CPT) that optimizes charging schedules for non-homogeneous heavy-duty fleets with multiple charging opportunities and grid constraints (R1, R2).
- A fleet management support-tool specification that defines interfaces between a Fleet Planning Tool, Energy Management System, Charging Station Management System, and a

	Vehicle Digital Twin (R3). - A battery digital twin method for predicting charging power profiles and charging time with low error for real charging sessions (R4). - Broader battery management expertise (State-of-X, Remaining Useful Life, diagnostics and (fast-)charging control) and a climate-controlled Battery Lab up to pack and vehicle level (R5, R6). - Megawatt charging readiness activities as a boundary condition for robustness validation, including vehicle digital twins for heavy-duty operations, in FlexMCS (R9). **Use cases** - Depot and corridor fleets where limited grid connection or peak constraints require scheduling (R1). - Smart energy hubs and megawatt charging sites where operational planning, battery health, and grid constraints must be co-optimized (R3, R9). - Truck monitoring and operational assessment frameworks, leveraging insights from MAGPIE electric truck demonstrations and deliverables (R10, R12). *Use cases involving megawatt charging reflect validation scenarios designed to stress-test scheduling algorithms and digital twin robustness under high-power conditions, rather than assumptions about near-term deployment of megawatt charging infrastructure.*
Rationale for Joint R&D Project	**Need and challenge** Scaling battery-electric fleets is constrained by charging infrastructure limits and grid congestion, making it difficult to ensure operational robustness and cost control (R1). Heavy-duty fleets add further complexity due to high energy demand and tight logistics, especially as megawatt charging systems are introduced (R9). **Why joint international R&D is required** - The Korean SME partner brings an existing charging-execution platform, expertise in relevant protocols and standards related to EV and charging, and market access in Korea. - TNO contributes validated model-based engineering methods and a toolchain combining testing, modelling, and optimal control, plus unique validation infrastructure (R5, R6, R7). - Joint international R&D is required to integrate TNO's validated model-based digital-twin and optimization methods with the Korean SME's charging management and execution platform, and to co-develop the missing integration layers (including a lightweight fleet charging planning interface and validation workflows) required to demonstrate TRL6 performance under grid-constrained, real-world conditions). **Strategic complementarity**

	- TNO provides algorithms and models: vehicle and battery digital twins, state estimation, charging-profile prediction, and optimization routines demonstrated in relevant research outputs (R2, R3, R4) and projects (R8, R9). - The SME provides integration into a scalable product platform (UI, deployment, V2X and virtual power plant interfaces where applicable), and access to operational datasets needed for calibration and validation. **Expected result of collaboration** A TRL6 demonstrator (prototype in relevant operational environment) that can be commercialized by the SME, with TNO contributing validated models, performance evidence, and a path to further standardization and international roll-out (R1, R3, R6).
Key Words	1. Charge planning 2. battery digital twin 3. vehicle digital twin 4. grid-constrained smart charging.

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (4) Prototype validated in laboratory or controlled pilot context (see TRL guidance in template; evidence base from R2, R4, R6).	**End of the Project**	Level (6) Prototype demonstrated in relevant operational environment (fleet depot or corridor charging site).
R&D Objectives	**O1.** Integrate a fleet charge planning engine with vehicle and battery digital twin services via standardized APIs suitable for integration with FMS, EMS, CSMS (build on interface concepts described in R3). Compatibility with OCPP 2.1 and ISO 15118-20 are considered as forward-looking, design-time requirements, but not mandatory pilot deployment conditions. *Fleet charge planning refers to the translation of basic operating schedules and travel intent into energy and charging constraints. A full fleet logistics planning system is not required; where such system exists, optional integration is supported.* **O2.** Develop and calibrate digital twin models for: - Vehicle energy consumption prediction for heavy-duty duty cycles (supporting route and operational variability). - Battery charging profile and charge-time prediction, and health-aware constraints (building on methods in R4 and TNO battery management capabilities in R5, R8). *In this context, the battery digital twin is limited to charging-behavior prediction and health-aware constraints consistent with R4, R5, and R8, and does not imply OEM-internal battery state estimation or absolute lifetime prediction.* **O3.** Extend the optimization objective to jointly minimize: - Peak grid power demand and grid limit violations (R1); treated as site-level and time-varying boundary conditions, - Electricity costs under dynamic tariffs, - Battery degradation proxies linked to operational charging decisions (R2, R5); relative decision penalties rather than absolute lifetime estimation. **O4.** Validate the integrated system through staged testing: - Offline simulation and scenario analysis, and - Hardware-in-the-loop or pack-level testing in TNO facilities where required (R6, R7), - Field pilot with the Korean SME partner fleet customers (target: TRL6). *High-power or megawatt-charging-level conditions, if used, should be treated as scenario-based robustness tests rather than assumptions about infrastructure deployment.*			

R&D Target Specifications	**Target performance indicators (project targets; to be validated during the project)** **S1. Charge-time and charging-profile prediction error**: - Robust and accurate charge time estimation and improved profile prediction. Target of <= 2% charge-time error compared to baseline heuristics; calibrated, vehicle specific prediction performance evaluated for a given vehicle-charger context using representative charging sessions, consistent with reported performance in R4 (to be re-validated for the partner fleet). **S2. Fleet scheduling computation:** - Target: compute feasible schedules for 50 vehicles, 20 chargers, and a 24h horizon within <= 5 minutes on a standard server (project target). **S3. Peak power reduction:** - Target: >= 15% reduction in peak power demand at the depot compared to uncontrolled charging (project target; motivation aligned with R1), without assuming participation in external grid services or dispatchable demand response. **S4. Energy cost reduction:** - Target: >= 5% reduction in energy cost with time-of-use tariffs (project target). **S5. Battery life impact:** - Target: demonstrable reduction in degradation proxy suitable for comparing charging strategies, supported by TNO battery management methods (R5, R8), rather than absolute lifetime prediction or warranty-related assessment. **Validation environments (capability statements)** - Module and pack testing with climatic control and bidirectional charge/discharge up to 750 V, 400 A, 300 kW in TNO facilities (R6). - Temperature ranges for climatic testing including -60°C to +150°C depending on facility (R6) and Battery Lab capabilities (R5). - Powertrains Test Centre (Helmond) facilities for heavy-duty relevant validation activities (R7).
R&D Plan	**Work plan (example 36-month program; can be adapted to TIPA scheduling)** **WP1 Requirements and data integration (M1-M6)** - Collect 1) operational requirements, 2) fleet duty-cycle data if available - otherwise simple schedules or rule-based policies, 3) charger and grid constraints from the Korean SME partner. - Define data model and interfaces for FMS, EMS, CSMS, and digital twins, aligning with interface concepts from Charging Energy Hubs tooling (R3).

Deliverables: Requirements specification; interface control document; anonymized pilot dataset.

WP2 Digital twin development (M4-M18)

- Vehicle digital twin: energy consumption prediction models for heavy-duty missions based on externally observable data, with uncertainty quantification.
- Battery digital twin: charging power profile and charge-time prediction services with adaptive parameter estimation, building on the approach in R4.
- Battery health layer: State-of-X and degradation-aware constraints, aligned with TNO battery management and diagnostics capabilities (R5, R8).

Deliverables: Digital twin services (API); calibration report; validation results on historical data.

WP3 Optimization and charge planning engine (M7-M24)

- Formulate scheduling problem: charger assignment, time windows, grid and tariff constraints, and degradation cost terms. *Optimization focuses on producing feasible and robust charging schedules, without assuming participation in external grid services or deterministic lifetime optimization.*
- Implement solver pipeline: mixed-integer optimization for discrete charger allocation plus model predictive control for rolling-horizon replanning (project design choice; method selection will be justified during WP1).
- Add robustness: handle delays, variable energy consumption, and stochastic arrivals.

Deliverables: Charge planning engine; robustness evaluation; integration guide.

WP4 Integration into SME product and V2X/VPP interfaces (M12-M30)

- Integrate TNO modules into the Korean SME platform, including user workflows, reporting, and optional V2X and virtual power plant coordination (architectural compatibility).
- Cybersecurity and data governance: implement secure data sharing and auditing features as required by partner deployments (project activity).

Deliverables: Integrated prototype; deployment documentation; operator dashboard.

WP5 Verification and demonstration (M18-M36)

- Lab validation where needed using TNO test facilities (R6, R7).
- Field pilot: demonstrate depot or corridor operation with the Korean SME partner, collecting KPIs for energy cost, peak reduction, schedule robustness, and battery health indicators.

	- Benchmark against baseline scheduling and charging heuristics. **Deliverables:** Pilot evaluation report; TRL6 demonstrator; exploitation plan; joint scientific publication(s) building on R2 and R4. **Evidence base and external learnings** - Use monitoring and operational evaluation practices from the MAGPIE e-truck demonstrations and deliverables to structure KPI collection and reporting (R10, R12). - Apply digital twin architectural principles consistent with modular DT definitions in MAGPIE deliverable D4.2 (R11).

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**TNO (Research Institute) will provide:** - System architecture for integrated vehicle charge planning and digital twin services, aligned with modular DT principles (R11) and CEH interface specifications (R3). - Algorithms and prototypes: - Charge Planning Tool concepts for commercial electric vehicles, including grid constraints and charger-vehicle compatibility (R1, R2). - Battery Digital Twin charging-profile prediction methods and implementation support (R4). - Battery management expertise: State-of-X, Remaining Useful Life, diagnostics, and charging control (R5, R8). - Heavy-duty energy modeling and vehicle digital twin expertise (R9). - Validation and testing resources: - Battery Lab and vehicle and battery test facilities including climatic testing and 300 kW charge/discharge capability (R5, R6). - Powertrains Test Centre resources for heavy-duty validation contexts and staff expertise (R7). - Dissemination and standardization: - Joint publications and contributions to relevant international networks through TNO and TU/e affiliations, including Steven Wilkins (R13).
Expected R&R of Korean SME Partner	**Korean SME Partner will provide:** - An existing charging management software platform for integration. - Productization capabilities: cloud deployment, operator UI, reporting, and customer support workflows. - Access to relevant operational data (charging sessions, charger metadata, tariffs, fleet-level data, without OEM-internal access) for model calibration and validation, with appropriate anonymization and data governance. - Pilot deployment environment in Korea: fleet operators and charging sites to demonstrate TRL6 performance. - Optional interfaces for V2X and virtual power plant participation where applicable to the partner business, to be co-designed in WP4. **Collaboration effort** - Joint definition of KPIs and validation protocols, including lessons learned from electric truck monitoring and operational demonstration practices (R10, R12).

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Expected technical outcomes** - Integrated prototype platform combining fleet charge planning and vehicle and battery digital twin services, delivered as modular services that can be embedded in commercial software (R3, R11). - Validated prediction services for charging power profile and charge time, extended and validated for the partner fleet conditions (R4). - Grid-aware, health-aware scheduling capability that can reduce peak power and improve operational robustness under limited grid connection, aligned with the use case motivations in R1. - Validation evidence and datasets (appropriately anonymized) demonstrating performance improvements compared to baseline charging strategies. **Expected impact** - Reduced operational barriers for electrifying heavy-duty fleets by enabling feasible charging schedules under constrained infrastructure (R1, R9). - Improved Total Cost of Ownership via coordinated scheduling and battery life extension strategies, consistent with TNO battery management focus (R5, R8). - Accelerated commercialization in Korea through the SME partner product platform, by validated robustness, reduced operational uncertainty, and improved decision confidence, supported by TNO validation evidence and test capabilities (R6, R7). - Knowledge outputs: at least one joint publication extending the existing research line of Beckers, Medina, and Wilkins (R2) and the battery DT work by Rath and Wilkins (R4). **Relevance to monitoring and demonstration practice** - The project will align KPI monitoring and reporting with public MAGPIE deliverables on electric heavy transport trucking demonstrations and digital tools, ensuring comparability and reproducibility of evaluation practices (R10, R12).

6. References

R1. TNO Insight: Optimal charging planning for logistics service providers considering grid congestion (5 Dec 2024). Accessed 2025-12-19. https://www.tno.nl/nl/newsroom/insights/2024/12/optimale-laadplanning-logistiek/

R2. Beckers, Medina, Wilkins. Charge Planning Tool for Heavy-Duty Electric Vehicle Fleets (conference paper, 2024). Accessed 2025-12-19. https://resolver.tno.nl/uuid:4043c8dc-e109-4441-8b23-0ea3a2fddf68

R3. Beckers et al. Charging Energy Hubs: Requirements and Specification of the Operational Fleet Management Support Tools (TNO report D2.4, 2025-02-28). Accessed 2025-12-19. https://publications.tno.nl/publication/34644650/KVx1RrIc/beckers-2025-D2.4charging.pdf

R4. Rath, Wilkins. Prediction of Electric Vehicle Charge Profile using Battery Digital Twin (2024). Accessed 2025-12-19. https://research.tue.nl/en/publications/prediction-of-electric-vehicle-charge-profile-using-battery-digit and PDF: https://research.tue.nl/files/346988394/Prediction_of_Electric_Vehicle_Charge_Profile_using_Battery_Digital_Twin_1_.pdf

R5. TNO web page: Sustainable batteries with advanced battery management (Battery Lab, State-of-X, RUL, diagnostics). Accessed 2025-12-19. https://www.tno.nl/en/sustainable/mobility-logistics/batteries/

R6. TNO brochure: Vehicle & Battery Testing (module and pack testing specs, climate chambers). Accessed 2025-12-19. https://publications.tno.nl/publication/34639492/gxMYkR/putte-2022-vehicle.pdf

R7. TNO web page: TNO research on powertrains (Powertrains Test Centre facilities). Accessed 2025-12-19. https://www.tno.nl/en/about-tno/organisation/units/mobility-built-environment/tno-research-powertrains/

R8. TNO news: Health of electric bus batteries now measurable on-site via charger (12 Apr 2025) and project VITALISE page. Accessed 2025-12-19. https://www.tno.nl/en/newsroom/2025/04/health-electric-bus-batteries/ ; https://www.tno.nl/en/sustainable/mobility-logistics/batteries/vitalise-battery-diagnostics-extension/

R9. TNO project page: FlexMCS (Flexible Megawatt Charging Systems), including vehicle digital twins and charging areas (project 1 Jan 2025 to 31 Dec 2028). Accessed 2025-12-19. https://www.tno.nl/nl/duurzaam/mobiliteit-logistiek/energiemanagement/flexmcs/

R10. MAGPIE project deliverables page (includes D6.2 heavy transport trucks demonstration report and D4.5 digital twin platforms and services). Accessed 2025-12-19. https://www.magpie-ports.eu/magpie-project/magpie-deliverables/

R11. MAGPIE Deliverable D4.2: Definition of Modular Architecture for Port Digital Twin (published 2023-05). Accessed 2025-12-19. https://www.magpie-ports.eu/wp-content/uploads/2023/05/Deliverable-D4.2-Definition-of-Modular-Architecture-for-Port-Digital-Twin.pdf

R12. TNO news: Demonstrations of automated driving and charging for logistics at Maasvlakte (MAGPIE, 10 Apr 2025). Accessed 2025-12-19. https://www.tno.nl/en/newsroom/2025/04/demo-automated-driving-charging-logistic/

R13. Steven Wilkins profile (TU/e research portal, noting Senior Research Scientist at TNO and BMS focus). Accessed 2025-12-19. https://research.tue.nl/en/persons/steven-wilkins/

R14. Beckers et al. A Modular Model for Determining Cornering Resistance of Heavy-Duty Vehicles (includes contact email for Alenka Beckers). Accessed 2025-12-19. https://research.tue.nl/files/346988176/A_Modular_Model_for_Determining_Cornering_Resistance_of_Heavy-Duty_Vehicles_1_.pdf

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization